TRIGONOMETRIC EQUATIONS

What is the first move?

Check the equation against each form in turn; the first match gives the first move.

1 A product equal to 0?
already factored

YES → **Set each factor to 0**
solve each basic equation, then take the union
→ Algebraic manipulation

NO

2 Two or more functions mixed?

YES → **Apply an identity**
Pythagorean, double-angle or another identity gives one function
→ Identity substitution

NO

3 Quadratic in a single function?

YES → **Substitute u = sin x**
factor or use the quadratic formula; reject u outside the range
→ Quadratic type

NO

4 A multiple angle nx?

YES → **Let u = nx**
solve in u, divide by n; up to 2n solutions on [0, 2π)
→ Multiple angles

NO

5 Needs squaring to reach one function?

YES → **Square, then check**
substitute every candidate into the original to discard extraneous values → Extraneous solutions

NO → **Basic equation**
check |a| against the range, find the reference angle, apply the quadrant rule → Basic sine equations

EVERY ROUTE ends at a basic equation sin x = a, cos x = a or tan x = a, then the general solution.